

Marco Martalò - Curriculum Vitae

Networks for Humans (Net4U) Lab
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PERSONAL INFORMATION

Place and date of birth: Galatina (LE), Italy, June 4, 1981
Nationality: Italian
Personal address: Via Piemonte 28, I-09060, Settimo San Pietro (CA), Italy
Personal phone: +39 320 9206519
Marital status: Married with two sons

WORKING EXPERIENCES

Associate Professor, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. December 2020 - present.

Post-Doc (*Assegnista di ricerca*) on “Efficient signal processing techniques,” Department of Engineering and Architecture, University of Parma, Italy. June 2017 - December 2020.

Assistant Professor (*Ricercatore a tempo determinato tipo A*), School of Engineering, E-Campus University, Novedrate (CO), Italy. May 2012 - April 2017. During the same period Research Associate at the Department of Information Engineering, University of Parma, Italy.

Post-Doc (*Assegnista di ricerca*) on “Efficient techniques for signal processing and transmission in wireless ad hoc and sensor networks,” Department of Information Engineering, University of Parma, Italy. February 2010 - April 2012.

SPINNER Research Fellow (*Borsista SPINNER*) on “Low-complexity localization in indoor scenarios,” Department of Information Engineering, University of Parma, Italy. January 2009 - January 2010. Tutor: Prof. G. Ferrari.

Research internship on “Network coding complexity,” School of Computer and Communication Sciences, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland. October 2007 - March 2008. Tutor: Prof. C. Fragouli.

EDUCATION

Ph.D. in Information Technologies, Department of Information Engineering, University of Parma, Italy, March 2009. Advisor: Prof. G. Ferrari.

Second Level Master of Science Degree (*Laurea specialistica*) Degree in Telecommunications Engineering, School of Engineering, University of Parma, Italy, December 2005. Grade: 110/110 *Summa Cum Laude*. Advisor: Prof. G. Ferrari.

First Level Bachelor of Science Degree (*Laurea triennale*) in Telecommunications Engineering, School of Engineering, University of Parma, Italy, September 2003. Grade: 103/110. Advisor: Prof. Riccardo Raheli.

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“Diploma” Degree, Liceo Classico “P. Colonna”, Galatina (LE), Italy, July 2000. Grade: 100/100.

RESEARCH INTERESTS

The activities carried out during my research career are in the general field of digital communication systems with particular attention to those in wireless environments. Scientific contributions are transversal to different types of systems, both traditional (point-to-point) and more advanced networking scenarios (with multiple users and ad-hoc communications). The results obtained can be classified into the following sub-topics (the first three active, the latter two not):

1. trustworthy and secure networking
2. advanced applications for communication networks;
3. audio signal processing;
4. transmission and signal processing techniques for sensor networks;
5. transmission systems on non-ideal channels.

DEVELOPED PROJECTS

Ongoing

“For active distribution Networks in Smart gridS (FITNESS),” project funded by the European Community under the Marie Skłodowska-Curie Actions - Doctoral Networks (MSCA-DN) scheme, 2024-2027. Principal investigator: Prof. E. Ghiani (University of Cagliari). **Prof. M. Martalò** acts as the coordinator of one of the Ph.D. candidates.

Past

“Monitoring and advanced security for optimized supply chain management (Safepath),” funded by the Centro di Competenza START 4.0 Ministry of Enterprises and Made in Italy under the National Recovery and Resilience Plan funded by the European Union - NextGenerationEU, 2024 - 2025. Principal Investigator: Prof. V. Pilloni (University of Cagliari).

“Virtualization and Orchestration of Liquid multi-Tenant network Architectures (VOLTA),” funded by the cascade call of the Spoke 6 of the project “RESearch and innovation on future Telecommunications systems and networks, to make Italy more smART (RESTART),” under the National Recovery and Resilience Plan funded by the European Union - NextGenerationEU, 2024 - 2025. Principal Investigator: **Prof. M. Martalò** (University of Cagliari).

“Smart Communities for Resilient Energy Transition (SCORET),” project funded by the Italian Ministry of University and Research (MUR) under the PRIN (Progetti di Ricerca di Interesse Nazionale) scheme, 2023-2025. Principal investigator: Prof. E. Ghiani (University of Cagliari).

“Security and RIghts In the CyBerSpace (SERICS) - spoke n. 3: in searCh Of eVidence of stEalth cybeR Threats (COVERT),” project n. PE00000014 funded by the Ministero dell’Università e Ricerca (MUR, Italy) under the National Recovery and Resilience Plan funded by the European Union - NextGenerationEU, 2023 - 2025. Principal Investigator: Prof. G. Giacinto (University of Cagliari).

“Ecosystem of Innovation for Next Generation Sardinia (e.INS) - spoke n. 6: Digital Transformation,” project n. ECS00000038 funded by the Ministero dell’Università e Ricerca (MUR, Italy) under the National Recovery and Resilience Plan funded by the European Union - NextGenerationEU, 2023 - 2025. Principal Investigator: Prof. G. Fenu (University of Cagliari).

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“POSition Information with Digital twin Offloading in trustworthy Next-generation Internet Applications (POSIDONIA),” funded by the third cascade call of the European project “NGI Sargasso,” 2024 - 2025. Principal Investigator: **Prof. M. Martalò** (University of Cagliari).

“Artificial Intelligence System Against Fare Evasion in the public transport (AI-SAFE),” regional project funded by Sardinia region, Italy, 2023 - 2025. Principal Investigator: **Prof. M. Martalò** (University of Cagliari).

“Active Control of the Acoustic Regions in a Vehicle,” collaboration between ASK Industries S.p.A., Reggio Emilia, Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2020 - 2023. Principal investigator: Prof. R. Raheli (University of Parma).

“Next generation smart perception sensors and distributed intelligence for proactive human monitoring in health, wellbeing, and automotive systems (NextPerception),” project n. 876487 funded by the European Commission and the Ministero dell’Istruzione, Università e Ricerca (MIUR, Italy) under the call H2020-ECSEL-2019-2-RIA-two-stage, 2020 - 2021. Prof. Martalò acted as lead writer of the deliverable D1.1 “State of the Art Assessment.” Principal Investigator: Prof. G. Ferrari (University of Parma).

“Airborne DATA Collection on Resilient System Architectures (ADACORSA),” project n. 876019 funded by the European Commission and the Ministero dell’Istruzione, Università e Ricerca (MIUR, Italy) under the call H2020-ECSEL-2019-2-RIA-two-stage, 2020 - 2021. Principal Investigator: Prof. G. Ferrari (University of Parma).

“Intelligent Secure Trustable Things (InSecTT),” project n. 876038 funded by the European Commission and the Ministero dello Sviluppo Economico (MISE, Italy) under the call H2020-ECSEL-2019-2-IA-two-stage, 2020 - 2021. Principal Investigator: Prof. G. Ferrari (Consorzio Interuniversitario Nazionale per l’Informatica, CINI, Research Unit of Parma).

“Control of the Acoustic Field in a Vehicle Cabin,” collaboration between ASK Industries S.p.A., Reggio Emilia, Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2018 - 2021. Principal investigator: Prof. R. Raheli (University of Parma).

“Cyber-Physical Systems for Safety in Production Plants with High Accident Risk with Integration of People/Objects Localization Technologies and Distributed Sensing Systems (CP-SEC),” collaboration between Istituto Nazionale contro gli Infortuni sul Lavoro (INAIL), Rome, Italy, the Department of Information Engineering and Mathematics, University of Siena, Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2020 - 2021. Principal investigators: Prof. A. Abrardo (University of Siena) and Prof. G. Ferrari (University of Parma).

“Intelligent Antennas for 5G Vehicular Communications,” collaboration between ASK Industries S.p.A., Reggio Emilia, Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2019 - 2021. Principal investigator: Prof. G. Ferrari (Consorzio Nazionale Interuniversitario per le Telecomunicazioni, CNIT, Research Unit of Parma, 2019-2020; University of Parma, 2020-2023).

“Acoustic Field Control in a Vehicle Cabin,” collaboration between ASK Industries S.p.A., Reggio Emilia, Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2018 - 2021. Principal investigator: Prof. R. Raheli (University of Parma).

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“Aggregate Farming in the Cloud” (AFarCloud),” project n. 783221 funded by the European Commission and the Ministero dell’Istruzione, Università e Ricerca (MIUR, Italy) under the call H2020-ECSEL-2017-2-RIA-two-stage, 2018 - 2021. Principal Investigator: Prof. G. Ferrari (University of Parma).

“Acoustic Field Control in a Tractor Cabin,” collaboration between Argo Tractors S.p.A., Fabbrico (RE), Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2018 - 2021. Principal investigator: Prof. A. Farina (University of Parma). Co-investigator: Prof. G. Ferrari.

“Prototype Design for Hybrid Radio/Inertial Navigation,” in collaboration with things2i S.r.l., Parma, Italy, 2019-2020. Principal investigator: Prof. G. Ferrari (University of Parma).

“Outdoor/Indoor Localization in 5G Networks,” collaboration between Huawei Ltd., Shanghai, China, and the Department of Engineering and Architecture, University of Parma, Italy, funded by the Huawei Innovation Research Program (HIRP), 2018 - 2020. Principal investigator: Prof. G. Ferrari (University of Parma).

“Analysis and Design of a Localization and Tracking Algorithm,” collaboration between Elettroc80 S.p.A., Viano (RE), Italy, and the Department of Engineering and Architecture, University of Parma, Italy, 2017 - 2019. Principal investigator: Prof. G. Ferrari (University of Parma).

“Advanced Digital Audio Processing Algorithms,” collaboration between CNIT, Parma, Italy, and MOVYM S.r.l., Milan, Italy, 2014. Principal investigator: Prof. G. Ferrari (CNIT, Research Unit of Parma).

“Phase Noise Suppression and Frequency Jumps Mitigation Algorithms,” collaboration between Huawei GmbH, Munich, Germany, and the Department of Information Engineering, University of Parma, Italy, 2012-2013. Principal investigator: Prof. R. Raheli (University of Parma).

“Cross-Network Effective Traffic Alerts Dissemination (X-NETAD),” joint Isreal-Italy project. Partners: Guglielmo Srl and the Department of Information Engineering, University of Parma (Italian side) and Cellint (Israeli side), 2011. Principal investigator: Prof. G. Ferrari (University of Parma).

“Design and Implementation of Advanced Vertical Handover Algorithms for HSDPA and Wi-Fi Networks,” collaboration between Guglielmo Srl, Reggio Emilia, Italy, and the Department of Information Engineering, University of Parma, Italy, 2010. Principal investigator: Prof. G. Ferrari (University of Parma).

“Algorithm for Audio Signature Detection,” collaboration between Elsas-Datamat, Rome, Italy, and the Department of Information Engineering, University of Parma, Italy, 2008-2009. Principal investigator: Prof. G. Ferrari (University of Parma).

“RFID-based System for Supply Chain Management,” collaboration between ID-Solutions, Parma, Italy, and the Department of Information Engineering, University of Parma, Italy, 2007. Principal investigator: Prof. G. Ferrari (University of Parma).

“Cooperative Remote Interconnected Measurement Systems Over Networks” (CRIMSON), Department of Information Engineering, University of Parma, Italy. PRIN (Progetti di Ricerca di Interesse Nazionale) project funded by the Italian Ministry of University and Research (MIUR), 2006-2008. Principal investigator: Prof. R. Raheli (University of Parma).

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AWARDS

National Scientific Qualification (*Abilitazione Scientifica Nazionale*) for Associate Professorship in Telecommunications, March 2018.

Elevation to IEEE Senior Member grade, December 2016.

First prize award, together with the WASNLab team, at the first Body Sensor Network (BSN) Contest, organized in conjunction with the 2011 Body Sensor Networks (BSN) conference, Dallas, TX, USA, May 2011.

Fulbright/BEST (Business Exchange Student Training) finalist, December 2006.

Best student paper award, *IEEE International Workshop on Wireless Ad hoc and Sensor Networks* (IWWAN), June 2006.

Award as best first-year student in Telecommunications Engineering (Second Level Master), University of Parma, Italy, for the academic year 2003-2004.

TUTORIALS, INVITED TALKS, AND SEMINARS

“Trustworthy by Design Next Generation Digital Twin-Enabled Services for Smart Environments,” keynote speech at the workshop “*Connected and Trustworthy Intelligence for Smart Environments: The Internet of Mirrors (IoM) and Beyond*” at the *IEEE WF-IoT* Abu Dhabi, United Arab Emirates, October 26, 2026.

“Trustworthy by Design Next Generation Digital Twin-Enabled Services,” invited online speech at the *University of Glasgow*, September 21, 2026.

“Distributed, Robust and Trustworthy Service Delivery in Edge-Assisted Green 6G,” tutorial with V. Pilloni and F. Malandra at the *IEEE Global Commun. Conf.* (Globecom), Taipei, Taiwan, December 12, 2025.

“Distributed, Robust and Trustworthy Service Delivery in Edge-Assisted Green 6G,” tutorial with V. Pilloni at the *IEEE Global Commun. Conf.* (Globecom), Cape Town, South Africa, December 12, 2024.

“Distributed, Robust and Trustworthy Service Delivery in Edge-Assisted Green IoT,” tutorial with V. Pilloni at the *IEEE World Forum on Internet of Things* (WF-IoT), Aveiro, Portugal, October 13, 2023. Held as virtual.

“Distributed, Robust and Trustworthy Service Delivery in Edge-Assisted Green 6G,” tutorial with V. Pilloni at the *IEEE Int. Symp. Personal, Indoor and Mobile Radio Commun.* (PIMRC), Toronto, Canada, September 5, 2023.

“On Fast and Secure Communications for IoT Networks,” *Int. Conf. Computing Technologies, Tools and Applications* (ICTAPP), University of Agriculture, Peshawar, Pakistan, May 10, 2023

“Distributed, Robust and Trustworthy Service Delivery in Edge-Assisted Green 6G,” tutorial with V. Pilloni at the *Int. Conf. Innovation in Clouds, Internet and Networks* (ICIN), Paris, France, March 7, 2023.

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“On Fast and Secure Communications for IoT Networks,” *EDUC Summer School on Building Trust in the Information Age*, University of Cagliari, Italy, August 30, 2022.

“Statistical Models of Spatially Correlated Binary Sources with Application to Communication Networks,” *International Conference on Advances in Multimedia (MMEDIA)*, Special Track on Models and Algorithms for Spatially and Temporally Correlated Data (STCD), Venice, Italy, April 27, 2017.

“Phase Noise Channel: Models and Fundamental Limits,” Institute for Communications Engineering, Technische Universität München (TUM), Germany. May 30, 2016.

“Towards Large-scale P2P Distributed Storage with Decentralized Maintenance: a Network Coding-based Approach,” Disney Research Zurich, Switzerland. December 9, 2011.

“Clustered Decentralized Binary Detection in Sensor Networks: a Joint Communication/Information-Theoretic Approach,” Department of Information Engineering, University of Parma, Italy. July 26, 2006.

SPEAKER AT CONFERENCES

2025 IEEE Global Communications Conference (Globecom 2025), Taipei, Taiwan, December 2025, to present the paper [C61].

2025 IEEE Conference on Communications (ICC 2025), Montreal, Canada, June 2025, to present the paper [C57].

2024 IEEE Global Communications Conference (Globecom 2024), Cape Town, South Africa, December 2024, to present the paper [C55].

2024 IEEE International Mediterranean Conference on Communications and Networking (MeditCom 2024), Madrid, Spain, July 2024, to present the paper [C53].

2019 IEEE International Conference on Computing Communication and Security (ICCCS 2019), Rome, Italy, October 2019, to present the paper [C40].

2019 International Symposium on Wireless Communication Systems (ISWCS 2019), Oulu, Finland, August 2019, to present the paper [C38].

2018 IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Bologna, Italy, September 2018, to present the paper [C37].

2016 International Symposium on Turbo Codes & Iterative Information Processing (ISTC 2016), Brest, France, September 2016, to present the paper [C36].

2015 International Symposium on Wireless Communication Systems (ISWCS 2015), Bruxelles, Belgium, August 2015, to present the papers [C33]-[C34].

2012 Riunione Annuale dell’Associazione Gruppo nazionale Telecomunicazioni e Tecnologie dell’Informazione (GTTI 2012), Cagliari & Villasimius, Italy, June 2012, to present the paper [CN5].

2010 IEEE International Symposium on Industrial Electronics (ISIE 2010), Bari, Italy, July 2010, to present the paper [C22].

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2010 International Symposium on Wireless Pervasive Computing (ISWPC 2010), Modena, Italy, May 2010, to present the papers [C19]-[C20].

2010 Information Theory and Applications Workshop (ITA 2010), San Diego, CA, USA, February 2010, to present the invited paper [C18].

2009 Riunione Annuale dell'Associazione Gruppo nazionale Telecomunicazioni e Tecnologie dell'Informazione (GTTI 2009), Parma, Italy, June 2009, to present the paper [CN3].

2008 Riunione Annuale dell'Associazione Gruppo nazionale Telecomunicazioni e Tecnologie dell'Informazione (GTTI 2008), Firenze, Italy, June 2008, to present the paper [CN2].

2008 IEEE International Symposium on Communications, Control and Signal Processing (ISCCSP 2008), St. Julians, Malta, March 2008, to present the papers [C9]-[C10].

2006 International Workshop on Wireless Ad-hoc Networks (IWWAN 2006), New York, NY, USA, June 2006, to present the paper [C4] awarded with a Best Student Paper Award.

SCOLARSHIPS

6-month scholarship on “Network coding complexity,” School of Computer and Communication Sciences, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland. October 2007.

1-year CNIT scholarship (out of 15 at a nation wide level) on the analysis and design of sensor networks. Assigned to the Department of Information Engineering, University of Parma, Italy. Not accepted for overlap with an Italian Ministry of University and Research (MIUR) scholarship. July 2006.

PRIN (nation-wide italian project) scholarship on “Mutual information and decentralized detection in clustered sensor networks,” Department of Information Engineering, University of Parma, Italy. Not accepted for overlap with an Italian Ministry of University and Research (MIUR) scholarship. March 2006.

Merit-based scholarships for completing undergraduate and graduate university studies (average grade above 24/30 in the undergraduate course and 28/30 in the graduate course). Academic years: 2000/2001, 2001/2002, 2002/2003, 2003/2004, 2004/2005, and 2005/2006.

TEACHING ACTIVITIES

Lecturer for *Traditional Courses*

“Network Security” (4 CFUs), University of Cagliari, Italy. Second-year course for the Master Degree in Internet Engineering. Academic years: from 2022-2023 to 2026-2027.

“Network Security Laboratory” (2 CFUs), University of Cagliari, Italy. Second-year course for the Master Degree in Internet Engineering. Academic years: from 2021-2022 to 2026-2027.

“Quantum Security” (2 CFUs), University of Cagliari, Italy. Second-year course for the Master Degree in Internet Engineering. Academic years: from 2025-2026 to 2026-2027.

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“Communication Networks” (6 CFUs), University of Cagliari, Italy. First-year course for the Bachelor Degree in Electrical, Electronic, and Computer Engineering. Academic years: from 2020-2021 to 2026-2027. Also held in a blended version until 2021-2022.

“Applications for Smart Cities” (4 CFUs), University of Cagliari, Italy. Second-year course for the Master Degree in Internet Engineering. Academic years: from 2020-2021 to 2021-2022.

“Internet and Multimedia” (6 CFUs), University of Parma, Italy. First-year course for the Bachelor Degree in Management Engineering. Academic year: 2019-2020 with delivery in the second semester with online teaching due to the COVID-19 pandemic.

“Internet and Multimedia” (9 CFUs), University of Parma, Italy. First-year course for the Bachelor Degree in Information Systems Engineering. Academic years: from 2017-2018 to 2019-2020.

“Communication Networks” (6 CFUs), University of Parma, Italy. Third-year course for the Bachelor Degree in Computer, Electronic, and Telecommunication Engineering and second-year course for the Bachelor Degree in Information Systems Engineering. Academic year: 2018-2019.

“Laboratory of Telecommunications (Wireless Networks)” (2 CFUs), University of Parma, Italy. Second-year course for the Master Degree in Telecommunications Engineering. Academic year: 2009-2010.

“Network Coding” (10 hours), University of Parma, Italy. Short course for the Ph.D. program in Information Technologies. February 2009.

Lecturer for *e-Learning* Courses

“Signals and Systems” (9 CFUs), E-Campus University, Italy. Second-year course for the Bachelor Degree in Computer and Automation Engineering. Academic years: from 2015-2016 to 2016-2017.

“Telecommunications” (9 CFUs), E-Campus University, Italy. Second-year course for the Bachelor Degree in Computer and Automation Engineering. Academic years: from 2012-2013 to 2016-2017.

“Communication Networks” (6 CFUs), E-Campus University, Italy. Third-year course for the Bachelor Degree in Computer and Automation Engineering. Academic years: from 2012-2013 to 2016-2017.

“Digital Signal Processing” (6 CFUs), E-Campus University, Italy. Third-year course for the Bachelor Degree in Computer and Automation Engineering. Academic years: from 2012-2013 to 2016-2017.

“Telecommunications and Remote Sensing” (6 CFUs), E-Campus University, Italy. Second-year course for the Master Degree in Computer and Automation Engineering. Academic years: from 2013-2014 to 2016-2017.

Teaching Assistant

Lectures, for a total amount of 32 hours, for the course of “Wireless Communications” (2018-2019) for the Master Degree in Communication Engineering at the University of Parma, Italy. For this course, I also served as chair of the examination committee.

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Integrative teaching activities (lectures, exercise and laboratory sessions, seminars, and tutoring) at the University of Parma, Italy, for a total amount of about 350 hours. Activities done for the following courses: “Network performance” (from 2015-2016 to 2019-2020), “Wireless Communications” (from 2016-2017 to 2019-2020), “Comunicazioni Wireless” (from 2010-2011 to 2012-2013), “Trasmissione numerica A” (from 2009-2010 to 2011-2012), “Teoria dei segnali B” (2008-2009), “Sistemi radiomobili e a larga banda” (from 2006-2007 to 2008-2009), “Percorso di alfabetizzazione informatica” (2006-2007).

Member of the examination committee for the following courses: “Wireless Communications” (from 2013-2015 to 2019-2020), “Comunicazioni wireless” (from 2010-2011 to 2012-2013), “Trasmissione numerica” (from 2011-2011 to 2013-2014), “Trasmissione numerica A” (from 2009-2010 to 2010-2011), “Sistemi radiomobili e a larga banda” (from 2007-2008 to 2008-2009).

Supervised Ph.D. Students

Oroos Arshi, Ph.D. in Electronic and Computer Engineering, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Research topic: “Advanced communication protocols in active distribution networks,” XLI cycle, 2025-2028 (in progress). **Advisor: Prof. M. Martalò.** Co-advisor: Prof. V. Pilloni.

Rahian Uddin, Ph.D. in Electronic and Computer Engineering, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Research topic: “Digital twin prosumer-aware energy communities,” XLI cycle, 2025-2028 (in progress). Advisor: Prof. V. Pilloni. **Co-advisor: Prof. M. Martalò.**

Giovanni Pettorru, Ph.D. in Electronic and Computer Engineering, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Energy-efficient and trustworthy location-based services for next-generation IoT,” XXXVIII cycle, 2022-2025 (defended in February 2026). Advisor: Prof. V. Pilloni. **Co-advisor: Prof. M. Martalò.** Partially presented work in the following publications: [J35], [J37], and [J39]-[J40].

Anatolij Borroni, Ph.D. in Information Technologies, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Active control of Personal Sound Zones by digital signal processing,” XXXVI cycle, 2020-2023 (defended in March 2023). Advisor: Prof. R. Raheli. Partially presented work in the following publications: [C44]-[C46] and [C48].

Alessandro Opinto, Ph.D. in Automotive for Intelligent Mobility, Department of Industrial Engineering, University of Bologna, Italy. Thesis title: “Control of the noise acoustic field in a vehicle cabin,” XXXIV cycle, 2018-2021 (defended in June 2022). Advisor: Prof. A. Farina. Partially presented work in the following publications: [J30].

Muhammad Asim, Ph.D. in Information Technologies, Department of Information Engineering, University of Parma, Italy. Thesis title: “Advanced receivers for next generation wireless communication systems,” XXVIII cycle, 2013-2015 (defended in March 2016). Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J25]-[J26].

Supervised Master Students (3+2 or 5-year program)

A. Coni, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Liquid digital twins: distributed architecture in the Edge-Fog-Cloud context,” November 2025. Advisor: Prof. V. Pilloni.

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G. Nurcis, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Sustainable energy data certification through Edge-IoT and Blockchain,” September 2025. **Advisor: Prof. M. Martalò.**

Alberto Mancosu, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Prototype and evaluation of an Edge-Based System for IoT Traffic Fingerprinting,” July 2025. **Advisor: Prof. M. Martalò.** Partially presented work in the following publication: [C63].

Matteo Flumini, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Encrypted network traffic classification using convolutional neural networks,” February 2023. **Advisor: Prof. M. Martalò.** Partially presented work in the following publication: [J39].

Giovanni Pettorru, Department of Electrical and Electronic Engineering, University of Cagliari, Italy. Thesis title: “Secure and low-latency communications based on WebSocket and QUIC in IoT scenarios,” *in Italian*, September 2022. **Advisor: Prof. M. Martalò.** Partially presented work in the following publication: [C47].

Riccardo Straccia, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Temporal convolutional networks for microphone virtualization,” March 2021. Advisor: Prof. R. Raheli.

Mona Shivaji Rao Chavan, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Synchronization issues in positioning systems for next-generation wireless systems,” December 2020. **Advisor: Dr. M. Martalò.**

Anatolij Borroni, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Performance analysis of digital signal processing algorithms for active control of individual acoustic zones,” October 2020. Advisor: Prof. R. Raheli. Partially presented work in the following publication: [C44]-[C46].

Gianmarco Carraglia, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Adaptive filter estimation for active noise control in a tractor cabin,” March 2020. **Advisor: Dr. M. Martalò.** Partially presented work in the following publication: [C45].

Fabrizio Carpi, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Exploring machine learning algorithms for decoding linear block codes,” October 2018. Advisor: Prof. R. Raheli. Partially presented work in the following publication: [C39].

Elia Santi, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Decoding Reed-Muller codes using minimum-weight parity checks,” October 2018. Advisor: Prof. R. Raheli.

Alessandro Opinto, Department of Engineering and Architecture, University of Parma, Italy. Thesis title: “Design and performance of a precoded OFDM massive MIMO system for new generation cellular network,” March 2018. Advisor: Prof. R. Raheli. Partially presented work in the following publication: [C37].

Antonino Gervasi, School of Engineering, University of Parma, Italy. Thesis title: “Information rate analysis of the oversampled phase-noise channel,” *in Italian*, March 2015. Advisor: Prof. R. Raheli. Partially presented work in the following publication: [C34].

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Carlo Tripodi, School of Engineering, University of Parma, Italy. Thesis title: “Information rate analysis of phase noise-limited communications,” December 2012. Advisor: Prof. R. Raheli. Partially presented work in the following publications: [J22] and [C31].

Michele Mohorovicich, School of Engineering, University of Parma, Italy. Thesis title: “Network-coded multihop multicast: topology and complexity,” March 2011. Advisor: Prof. G. Ferrari. Partially presented work in the following publication: [C29].

Naldo Poletti, School of Engineering, University of Parma, Italy. Thesis title: “Design and experimental development of a wireless network with mesh topology,” *in Italian*, March 2011. Advisor: Prof. G. Ferrari

Erind Meco, School of Engineering, University of Parma, Italy. Thesis title: “Design of a network-coded P2P architecture for managing large information flows,” *in Italian*, March 2011. Advisor: Prof. M. Amoretti. Partially presented work in the following publication: [C30].

Davide Ribolini, School of Engineering, University of Parma, Italy. Thesis title: “Highly energy efficient target tracking in clustered sensor networks,” *in Italian*, December 2010. Advisor: Prof. G. Ferrari.

Matteo Giuberti, School of Engineering, University of Parma, Italy. Thesis title: “Design of localization algorithms for motion capture in wireless SunSpot sensor networks,” *in Italian*, December 2010. Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J19] and [C26].

Giovanni Spigoni, School of Engineering, University of Parma, Italy. Thesis title: “Design and performance analysis of a software for vertical handover between heterogeneous networks,” *in Italian*, July 2010. Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J9] and [C23].

Riccardo Bussandri, School of Engineering, University of Parma, Italy. Thesis title: “Design and performance evaluation of a P2P distributed storage system based on network and erasure coding,” *in Italian*, December 2009. Advisor: Prof. M. Amoretti. Partially presented work in the following publication: [C21].

Stefano Busanelli, School of Engineering, University of Parma, Italy. Thesis title: “Markov chain models for performance analysis of sensor networks with multihop communications,” *in Italian*, December 2007. Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J5] and [C9], [C11], [C17].

Marco Sarti, School of Engineering, University of Parma, Italy. Thesis title: “Algorithms for Distributed Detection of Non-constant Binary Phenomena in Sensor Networks,” *in Italian*, December 2006. Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J7] and [C6].

Paolo Medagliani, School of Engineering, University of Parma, Italy. Thesis title: “Design and Implementation of Wireless Sensor Networks with Zigbee Technology,” *in Italian*, April 2006. Advisor: Prof. G. Ferrari. Partially presented work in the following publications: [J1] and [C7].

Marco Martalò - Curriculum Vitae

Supervised Bachelor Students (3-year program)

A total amount of 36 Bachelor students have been supervised, 25 of them as advisor and the remaining 11 as co-advisor. The full list is available on the personal page <https://marcomartalo.github.io>

PROFESSIONAL ACTIVITIES

Journal Editor and Conference Organization

Associate Editor for IEEE Networking Letters journal, January 2026-present. Coordination of the review process for an average amount of 1 paper per month.

Associate Editor for IEEE Communications Letters journal, November 2025-present. Coordination of the review process for an average amount of 1 paper per month.

Associate Editor for IEEE Access journal, June 2018-present. Coordination of the review process for an average amount of 4 papers per month.

Guest Editor for the Special Issue on “IoT Enabling Technologies for Smart Cities: Challenges and Approaches,” MDPI Sensors, 2021-2024.

Moderator for the IEEE TechRxiv preprint platform, April 2020-present. Coordination of the approval process for an average amount of 2.5 papers per month.

Topic Advisory Panel Member for the MDPI Sensors journal, August 2020-present.

General Chair of the IEEE International Mediterranean Conference on Communications and Networking (MeditCom), Cagliari, Italy, 2026.

Track Chair for “Selected Areas in Communications: e-Health” of the IEEE International Conference on Communications (ICC), Washington DC, USA, 2027.

Track Chair for “Secure and Scalable Network Architectures” of the IEEE Middle East Conference on Communications and Networking (MECOM), Cairo, Egypt, 2025.

Publicity Chair of the IEEE Latin American Conference on Internet of Things (LatinIoT), Santiago, Chile, 2027.

Publicity Chair of the IEEE World Forum on Internet of Things (WF-IoT), Abu Dhabi, United Arab Emirates, 2026.

Publicity Chair of the IEEE Conference on Communications and Network Security (CNS), Avignon, France, 2025.

Publication Chair of the IEEE IoT Vertical and Topical Summit on Tourism, Cagliari, Italy, 2021.

Marco Martalò - Curriculum Vitae

Organizer for the following Special Sessions.

- “Next-Generation Networking and Computing Continuum for Large-Scale Digital Twins” special session in the International Symposium on Distributed Simulation and Real Time Applications (DS-RT), Padua, Italy, 2026;
- “Wireless Intelligence: From Reconfigurable Surfaces to Edge/Cloud Communications” workshop of the Gruppo nazionale Telecomunicazioni e Teoria dell’Informazione (GTTI), online, 26 March 2021;
- “Recent Advances in Indoor Navigation for IoT-based Applications” (RAIN-IoT) in the International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Bologna, Italy, 2018.
- “STCD: Models and Algorithms for Spatially and Temporally Correlated Data” in the International Conference on Advances in Multimedia (MMEDIA), Venice, Italy, 2017.

TPC member for the following conferences.

- IEEE International Conference on Communications (ICC), Communication Theory Symposium (2018-2027), IoT & Sensor Networks (2026-2027), Mobile and Wireless Networks Symposium (2022-2027), Next-Generation Networking and Internet Symposium (2025-2027), Communication QoS, Reliability & Modeling Symposium (2025-2027), Wireless Communications Symposium (2024-2027), Selected Areas in Communications E-Health track (2026), and Selected Areas in Communications Integrated Sensing and Communication track (2026-2027);
- IEEE Global Communications Conference (GLOBECOM), Communication Theory Symposium (2011,2026), IoT and Sensor Networks Symposium (2025-2026), Mobile and Wireless Networks Symposium (2024-2026), Next-Generation Networking and Internet Symposium (2026), and Wireless Communication Symposium (2021), Selected Areas in Communications E-Health track (2024-2026), Selected Areas in Communications Integrated Sensing and Communication track (2026), and Selected Areas in Communications Machine Learning for Communications and Networking track (2026);
- IEEE Conference on Communications and Network Security (CNS), 2025-2026;
- International Conference on Network of the Future (NoF), 2023-2026;
- International Conference on Future Communications and Networks (FCN), 2026;
- IEEE Future Networks World Forum (FNWF), 2026;
- International Conference on Wireless Communications and Signal Processing (WCSP), 2026;
- IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), (2020,2023-2024);
- International Conference on Advances in Mobile Computing & Multimedia Intelligence (MoMM), 2023-2024;
- International Conference on Future Internet of Things and Cloud (FiCloud), 2022-2024.
- International Conference on Advances in Multimedia (MMEDIA), 2018-2020.
- IEEE International Symposium on Power Line Communications and its Applications (ISPLC), 2020.
- IEEE International Conference on Computing Communication and Security (ICCCS), 2019.
- Workshop on Dependable Communication and Localization for the IoT, Graz, Austria, 2017.
- International Conference on Recent Advances in Electronics and Communication Technology (ICRAECT), Bengaluru, Karnataka, India, 2017.
- International Conference on Advances in Satellite and Space Communications (SPACOMM), 2009-2010.
- International Workshop on Performance Methodologies and Tools for Wireless Sensor Networks (WSNPerf), Pisa, Italy, 2009.

Marco Martalò - Curriculum Vitae

Chair for the following conference sessions.

- “RAIN-IoT: Recent Advances in Indoor Navigation for IoT-based Applications,” International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Bologna, Italy, September 2018.
- “STCD: Models and Algorithms for Spatially and Temporally Correlated Data,” International Conference on Advances in Multimedia (MMEDIA), Venice, Italy, April 2017.
- “SPCI - Dialog 1: Electric Machines / Signal Processing,” IEEE International Symposium on Industrial Electronics (ISIE), Bari, Italy, July 2010.

Frequent and regular reviewer for the main scientific journals and conferences in the field of telecommunications. The full list is available on the personal page <https://marcomartalo.github.io>

Other Academic Duties

Secretary of the Italian Chapter of the IEEE Communications Society. 2026-2027.

Representative for the University of Cagliari at the Datacom Industry Association (DIA). 2025 - ongoing.

Representative of the research unity of GTTI at the University of Cagliari, Italy. 2024 - ongoing.

Committee member for drafting the degree program, Master in Internet Engineering, University of Cagliari, Italy. 2023 - ongoing.

Committee member for evaluating the students’ study plans, Bachelor in Electrical Energy Engineering for Sustainable Development, University of Cagliari, Italy. 2023 - ongoing.

Dean’s delegate for shareholder’s assembly, Consorzio Nazionale Interuniversitario per le Telecomunicazioni (CNIT), Italy. 2023 - ongoing.

Member of the scientific committee for the national GTTI meeting, Italy, 2021.

Co-chair for the annual PhD Day of the PhD Program in Electronic and Computer Engineering (DRIEI), Department of Electrical and Electronic Engineering, University of Cagliari, Italy, 2021.

Committee member for drafting the review report (*Rapporto del Riesame*), School of Engineering, E-Campus University, Italy. Academic years: from 2012-2013 to 2016-2017.

Committee member for quality assurance, School of Engineering, E-Campus University, Italy. Academic years: from 2013-2014 to 2016-2017.

PERSONAL SKILLS AND INTERESTS

Languages: English (good), French (basic), and German (basic).

Main personal interest as volleyball player and trainer (second-level Italian degree).

LIST OF PUBLICATIONS

Bibliometric indexes

source	citations	<i>h</i> -index
Google Scholar	1373	18
Scopus	840	16

Books

[B2] G. Boato, F. Gringoli, and **M. Martalò** (Eds.), “CNIT Technical Report-11: Towards a Trustworthy Information Exchange in the Digital Era,” Texmat, Italy, 2023. ISBN: 978-8-894-98272-5.

[B1] C. Buratti, **M. Martalò**, R. Verdone, and G. Ferrari, “Sensor Networks with IEEE 802.15.4 Systems: Distributed Processing, MAC, and Connectivity,” Springer, Germany, 2011. ISBN: 978-3-642-17489-6.

Book Chapters

[BC10] V. Pilloni, **M. Martalò**, and L. Atzori, “Trustworthy Task Allocation in IoT: a Cognitive Game-Theoretical Use Case,” chapter contribution in *CNIT Technical Report-11: Towards a Trustworthy Information Exchange in the Digital Era*, pp. 191-205, edited by G. Boato, F. Gringoli, and M. Martalò. Texmat, Rome, Italy, 2023. ISBN: 978-8-894-98272-5.

[BC9] **M. Martalò**, V. Pilloni, T. F. Rahman, and L. Atzori, “Game Theoretical-Based Task Allocation in Malicious Cognitive IoT,” chapter contribution in *Wireless Mesh Networks for IoT and Smart Cities: Technologies and Applications*, pp. 247-264, edited by L. Davoli and G. Ferrari. IET, London, UK, 2022. ISBN: 978-1-83953-282-5.

[BC8] L. Davoli, V. Mattioli, S. Gambetta, L. Belli, L. Carnevali, **M. Martalò**, A. Sgoifo, R. Raheli, and G. Ferrari, “Non-Invasive Psycho-Physiological Driver Monitoring Through IoT-Oriented Systems,” chapter contribution in *The Internet of Medical Things: Enabling Technologies and Emerging Applications*, pp. 19-33, edited by S. Kumar Pani, P. Patra, G. Ferrari, R. Kraveva, and H. Wang. IET, London, UK, 2021. ISBN: 978-1-83953-273-3.

[BC7] A. Abrardo, **M. Martalò**, and G. Ferrari, “Decision Fusion in Cognitive Wireless Sensor Networks,” chapter contribution in *Multisensor Data Fusion: From Algorithms and Architectural Design to Applications*, pp. 349-362, edited by H. Fourati. CRC Press, 2015. ISBN: 978-1-4822-6374-9.

[BC6] G. Spigoni, S. Busanelli, **M. Martalò**, G. Ferrari, and N. Iotti, “Vertical Handover in Heterogeneous Networks: a Comparative Experimental and Simulation-based Investigation,” chapter contribution in *Heterogeneous Cellular Networks*, pp. 265-286, edited by R. Q. Hu and Y. Qian. Wiley, 2013. ISBN: 978-1-1199-9912-6.

[BC5] **M. Martalò**, G. Ferrari, and C. Malavenda, “Wireless Sensor Networks and Audio Signal Processing for Homeland Security,” chapter contribution in *Effective Surveillance for Homeland Security: Balancing Technology and Social Issues*, pp. 457-488, edited by F. Flammini, R. Setola, and G. Franceschetti. Chapman and Hall/CRC Press (Taylor and Francis Group), 2013. ISBN: 978-1-4398-8324-2.

Marco Martalò - Curriculum Vitae

[BC4] S. Busanelli, **M. Martalò**, G. Ferrari, G. Spigoni, and N. Iotti, “Experimental Investigation of Vertical Handover Algorithms between WiFi and UMTS Networks,” chapter contribution in *Communication and Networking, Part I*, pp. 137-146, edited by C.-C. Chang, M. Li, C. Rong, C. Z. Patrikakis, and D. Slezak. Springer, 2011. ISBN: 978-3642175862.

[BC3] **M. Martalò** and G. Ferrari, “Low-complexity Audio Signal Processing for Localization in Indoor Scenarios,” chapter contribution in *The Internet of Things: 20th Tyrrhenian Workshop on Digital Communications*, pp. 167-176, edited by D. Giusto, A. Iera, G. Morabito, and L. Atzori. Springer, 2010. ISBN: 978-1-4419-1673-0.

[BC2] G. Ferrari, **M. Martalò**, and M. Sarti, “Reduced-Complexity Decentralized Detection of Spatially Non-constant Phenomena,” chapter contribution in *Grid Enabled Instrumentation and Measurement*, pp. 33-54, edited by F. Davoli, N. Meyer, R. Pugliese, and S. Zappatore. Springer, October 2008. ISBN: 978-0-387-09662-9.

[BC1] G. Ferrari, P. Medagliani, and **M. Martalò**, “Performance Analysis of Zigbee Wireless Sensor Networks with Relaying,” chapter contribution in *Grid Enabled Instrumentation and Measurement*, pp. 55-79, edited by F. Davoli, N. Meyer, R. Pugliese, and S. Zappatore. Springer, October 2008. ISBN: 978-0-387-09662-9.

Journal Papers

[J43] G. Pettorru, V. Pilloni, **M. Martalò**, and S. I. Lopes, “A multimodal RSSI-based dataset for indoor navigation in interference-prone environments,” in *Elsevier Future Generation Computer Systems*, vol. 184, November 2026.

[J42] A. Borroni, **M. Martalò**, A. Costalunga, C. Tripodi, and R. Raheli, “Stochastic modeling of car cabin acoustics with application to personal sound zone systems,” *Elsevier Applied Acoustics*, vol. 251, July 2026.

[J41] G. Pettorru, V. Pilloni, and **M. Martalò**, “Robust range-based localization approaches leveraging multiple wireless interfaces,” in *IEEE Trans. on Wireless Commun.*, vol. 25, pp. 20544-20557, 2026.

[J40] G. Pettorru and **M. Martalò**, “A persistent and secure publish-subscriber architecture for low-latency IoT communications,” in *IEEE Trans. on Network and Service Management*, vol. 23, pp. 729-739, 2026.

[J39] G. Pettorru, M. Flumini, and **M. Martalò**, “Balancing complexity and performance in convolutional neural network models for QUIC traffic classification,” *MDPI Sensors*, Special Issue on “Integrating Cloud Computing, Sensing Technologies, and Digital Twins in IoT Applications,” vol. 25, no. 15, July 2025.

[J38] E. Tuyishime, **M. Martalò**, P.A. Cotfas, V. Popescu, D. T. Cotfas, and A. Rekeraho, “Resource-efficient traffic classification using feature selection for Message Queuing Telemetry Transport-Internet of Things network-based security attacks,” *MDPI Applied Sciences*, Special Issue on “Advances in Internet of Things (IoT) Security: Challenges and Applications,” vol. 15, no. 8, April 2025.

[J37] **M. Martalò**, G. Pettorru, and L. Atzori, “A cross-layer survey on secure and low-latency communications in next-generation IoT,” in *IEEE Trans. on Network and Service Management*, vol. 21, no. 4, pp. 4669-4685, August 2024.

Marco Martalò - Curriculum Vitae

- [J36] A. Opinto, **M. Martalò**, R. Straccia, and R. Raheli, “An experimental performance assessment of temporal convolutional networks for microphone virtualization in a car cabin,” *MDPI Sensors*, Topical Collection on “*Wireless Sensor Networks Towards the Internet of Things*,” vol. 24, no. 16, August 2024.
- [J35] G. Pettorru, V. Pilloni, and **M. Martalò**, “Trustworthy localization in IoT networks: a survey of localization techniques, threats, and mitigation,” *MDPI Sensors*, Topical Collection on “*Wireless Sensor Networks Towards the Internet of Things*,” vol. 24, no. 7, March 2024.
- [J34] M. Amadeo, **M. Martalò**, G. Ruggeri, and M. Nitti, “Enabling social digital twins in the 6G era with information centric networking,” in *IEEE Communications Magazine*, vol. 61, no. 12, pp. 112-117, December 2023.
- [J33] **M. Martalò**, S. Perri, G. Verdano, F. De Mola, F. Monica, and G. Ferrari, “Hybrid UWB-inertial TDoA-based target tracking with concentrated anchors,” *IEEE Internet of Things Journal*, vol. 10, no. 14, pp. 12676-126895, July 2023.
- [J32] F. Carpi, **M. Martalò**, L. Davoli, A. Cilfone, Y. Yu, Y. Wang, and G. Ferrari, “Experimental analysis of RSSI-based localization algorithms with NLOS pre-mitigation for IoT applications,” *Elsevier Computer Networks*, vol. 225, pp. 1-14, April 2023.
- [J31] **M. Martalò** and R. Raheli, “Outage capacity analysis of the massive MIMO diversity channel,” *Elsevier Physical Communications*, vol. 53, pp. 1-9, August 2022.
- [J30] A. Opinto, **M. Martalò**, A. Costalunga, N. Strozzi, C. Tripodi, and R. Raheli, “Experimental analysis and design guidelines for microphone virtualization in automotive scenarios,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 2337-2346, 2022.
- [J29] **M. Martalò**, S. Perri, G. Verdano, F. De Mola, F. Monica, and G. Ferrari, “Improved UWB TDoA-based positioning using a single hotspot for industrial IoT applications,” *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3915-3925, June 2022.
- [J28] L. Davoli, **M. Martalò**, A. Cilfone, L. Belli, G. Ferrari, R. Presta, R. Montanari, M. Mengoni, L. Giraldo, E. G. Amparore, M. Botta, I. Drago, G. Carbonara, A. Castellano, and J. Plomp, “On driver behavior recognition for increased safety: a roadmap,” *MDPI Safety*, Special Issue on “*Traffic Safety and Driver Behaviour*,” vol. 6, n. 4, December 2020.
- [J27] A. Abrardo, **M. Martalò**, and G. Ferrari, “Information fusion for efficient target detection in large-scale surveillance Wireless Sensor Networks,” *Elsevier Information Fusion*, Special Issue on “*Event-Based Distributed Information Fusion Over Sensor Networks*,” vol. 38, pp. 55-64, November 2017.
- [J26] **M. Martalò**, G. Ferrari, M. Asim, J. Gambini, C. Mazzucco, G. Cannalire, S. Bianchi, and R. Raheli, “Iterative synchronization for dually-polarized independent transmission streams,” *IEEE Transactions on Communications*, vol. 65, n. 6, pp. 2534-2542, June 2017.
- [J25] **M. Martalò**, G. Ferrari, M. Asim, J. Gambini, C. Mazzucco, G. Cannalire, S. Bianchi, and R. Raheli, “Pragmatic phase noise compensation for high-order coded modulations,” *IET Communications*, vol. 10, n. 15, pp. 1956-1963, October 2016.

Marco Martalò - Curriculum Vitae

- [J24] A. Abrardo, **M. Martalò**, and G. Ferrari, "Impact of the knowledge of nodes' positions on spectrum sensing strategies in cognitive networks," *Elsevier Physical Communications*, Special Issue on "Self-Optimizing Cognitive Radio Technologies," vol. 19, pp. 84-92, June 2016.
- [J23] **M. Martalò** and R. Raheli, "Models, statistics, and rates of binary correlated sources," *Elsevier Physical Communications*, vol. 19, pp. 70-80, June 2016.
- [J22] **M. Martalò**, C. Tripodi, and R. Raheli, "Simple upper bound on the information rate of the phase noise channel," *IET Electronics Letters*, vol. 52, n. 7, pp. 517-519, April 2016.
- [J21] A. Gorrieri, **M. Martalò**, S. Busanelli, and G. Ferrari, "Clustering and sensing with decentralized detection in vehicular ad hoc networks," *Elsevier Ad Hoc Networks*, Special Issue on "Vehicular Networking for Mobile Crowd Sensing," vol. 36, part 2, pp. 450-464, January 2016.
- [J20] M. Picone, M. Amoretti, **M. Martalò**, F. Zanichelli, and G. Ferrari, "Combining geo-referencing and network coding for distributed large scale information management," *Wiley Concurrency and Computation: Practice and Experience*, Special Issue on "Advances in High Performance Computing and Simulation," vol. 27, n. 13, pp. 3295-3315, September 2015.
- [J19] M. Giuberti, **M. Martalò**, and G. Ferrari, "A hybrid radio/accelerometric approach to arm posture recognition," *Journal of Ambient Intelligence and Smart Environments*, vol. 7, n. 4, pp. 563-578, July 2015.
- [J18] A. Abrardo, G. Ferrari, **M. Martalò**, M. Franceschini, and R. Raheli, "Orthogonal multiple access with correlated sources: achievable region and pragmatic schemes," *IEEE Transactions on Communications*, vol. 62, n. 7, pp. 2531-2543, July 2014.
- [J17] **M. Martalò**, M. Amoretti, M. Picone, and G. Ferrari, "Sporadic decentralized resource maintenance for P2P distributed storage networks," *Elsevier Journal of Parallel and Distributed Computing*, vol. 74, n. 2, pp. 2029-2038, February 2014.
- [J16] G. Ferrari, **M. Martalò**, and A. Abrardo, "Information fusion in wireless sensor networks with source correlation," *Elsevier Information Fusion*, vol. 15, pp. 80-89, January 2014.
- [J15] **M. Martalò**, C. Buratti, G. Ferrari, and R. Verdone, "Clustered IEEE 802.15.4 sensor networks with data aggregation: energy consumption and probability of error," *IEEE Wireless Communications Letters*, vol. 2, no. 1, pp. 70-73, February 2013.
- [J14] **M. Martalò**, G. Ferrari, and C. Malavenda, "Low-complexity hybrid time-frequency audio signal pattern detection," *IEEE Sensors Journal*, vol. 13, no. 2, pp. 501-509, January 2013.
- [J13] **M. Martalò** and G. Ferrari, "Decoding and fusion in distributed detection schemes with unreliable communications," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 48, no. 1, pp. 16-26, January 2012.
- [J12] P. Medagliani, **M. Martalò**, and G. Ferrari, "Clustered Zigbee networks with data fusion: characterization and performance analysis," *Elsevier Ad Hoc Networks*, vol. 9, no. 7, pp. 1083-1103, September 2011.
- [J11] A. Abrardo, G. Ferrari, and **M. Martalò**, "On non-cooperative block-faded orthogonal multiple access schemes with correlated sources," *IEEE Transactions on Communications*, vol. 59, no. 7, pp. 1916-1926, July 2011.

Marco Martalò - Curriculum Vitae

- [J10] G. Ferrari, **M. Martalò**, and R. Pagliari, “Decentralized detection in clustered sensor networks,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 47, no. 2, pp. 959-973, April 2011.
- [J9] S. Busanelli, **M. Martalò**, G. Ferrari, G. Spigoni, and N. Iotti, “Vertical handover between WiFi and UMTS networks: experimental performance analysis,” *International Journal of Energy, Information and Communications*, vol. 2, no. 1, pp. 75-96, February 2011.
- [J8] **M. Martalò**, C. Buratti, G. Ferrari, and R. Verdone, “Decentralized detection in IEEE 802.15.4 wireless sensor networks,” *EURASIP Journal on Wireless Communication and Networking*, Special Issue on “*Signal Processing-assisted Protocols and Algorithms for Cooperating Objects and Wireless Sensor Networks*,” vol. 2010, Article ID 174063, 10 pages, 2010.
- [J7] **M. Martalò** and G. Ferrari, “Low-complexity one-dimensional edge detection in wireless sensor networks,” *EURASIP Journal on Wireless Communication and Networking*, Special Issue on “*Signal Processing-assisted Protocols and Algorithms for Cooperating Objects and Wireless Sensor Networks*,” vol. 2010, Article ID 751520, 13 pages, 2010.
- [J6] **M. Martalò** and G. Ferrari, “A simple information-theoretic analysis of clustered sensor networks with decentralized detection,” *IEEE Communications Letters*, vol. 14, no. 6, pp. 560-562, June 2010.
- [J5] **M. Martalò**, S. Busanelli, and G. Ferrari, “Markov chain-based performance analysis of multi-hop IEEE 802.15.4 wireless networks,” *Elsevier Performance Evaluation (PEVA)*, Special Issue on “*Performance Evaluation of Wireless Ad Hoc, Sensor, and Ubiquitous Networks*,” vol. 66, no. 12, pp. 722-741, December 2009.
- [J4] A. Abrardo, G. Ferrari, **M. Martalò**, and F. Perna, “Feedback power control strategies in wireless sensor networks with joint channel decoding,” *MDPI Sensors*, Special Issue on “*Wireless Sensor Technologies and Applications*,” vol. 9, no. 11, pp. 8776-8809, November 2009.
- [J3] G. Ferrari, R. Pagliari, and **M. Martalò**, “Decentralized binary detection with non-constant SNR profile at the sensors,” *International Journal on Sensor Networks*, Special Issue on “*Energy-Efficient Algorithm and Protocol Design in Sensor Networks*,” vol. 4, nos. 1-2, pp. 23-36, 2008.
- [J2] G. Ferrari and **M. Martalò**, “Extending the lifetime of sensor networks through adaptive reclustering,” *EURASIP Journal on Wireless Communication and Networking*, Special Issue on “*Novel Techniques for Analysis and Design of Cross-Layer Optimized Wireless Sensor Networks*,” vol. 2007, Article ID 31809, 20 pages, 2007.
- [J1] G. Ferrari, P. Medagliani, S. Di Piazza, and **M. Martalò**, “Wireless sensor networks: performance analysis in indoor scenarios,” *EURASIP Journal on Wireless Communication and Networking*, Special Issue on “*MobileMAN (Mobile Multi-hop Ad Hoc Networks): From Theory to Reality*,” vol. 2007, Article ID 81864, 14 pages, 2007.

Conference Papers

- [C67] G. Pettorru, M. Curridori, **M. Martalò**, and V. Pilloni, “On the feasibility and Byzantine resilience of WiFi HaLow fingerprinting-based IoT positioning,” *Proc. IEEE World Forum on Internet of Things*, Abu Dhabi, United Arab Emirates, October 2026.

Marco Martalò - Curriculum Vitae

- [C66] G. Nurchis, A. Coni, G. Pettorru, V. Pilloni, and **M. Martalò**, “Lightweight and tamper-evident: a decentralized certification pipeline for the Internet of Energy,” *Proc. IEEE World Forum on Internet of Things*, Abu Dhabi, United Arab Emirates, October 2026.
- [C65] F. Falchi, G. Pettorru, V. Pilloni, and **M. Martalò**, “A feasibility study of Edge-centric digital twin for autonomous agricultural decision support,” *Proc. IEEE Conf. on AgriFood Electronics (CAFE)*, Cagliari, Italy, September 2026.
- [C64] A. Coni **M. Martalò**, and V. Pilloni, “Liquid Digital Twins: distributed architecture in the Edge-Fog-Cloud context,” *Proc. IEEE Int. Conf. Commun. (ICC)*, Glasgow, Scotland, May 2026.
- [C63] A. Mancosu, G. Pettorru, and **M. Martalò**, “FiSH: Feature-informed Similarity Hashing for Edge-based traffic fingerprinting,” *Proc. IEEE Int. Conf. Commun. (ICC)*, Glasgow, Scotland, May 2026.
- [C62] F. Marcello, D. Petza, **M. Martalò**, and V. Pilloni, “Digital twin-based framework for behavioral modeling and activity recognition in smart homes,” *Proc. IEEE Int. Conf. Commun. (ICC)*, Glasgow, Scotland, May 2026.
- [C61] A. Y. Chouquir, S. Saba, C. C. González, **M. Martalò**, and V. Pilloni, “A quality-driven network digital twin model with application to energy efficient mobile networks,” *Proc. IEEE Global Commun. Conf. (Globecom)*, Taipei, Taiwan, December 2025.
- [C60] L. Pisu, G. Pettorru, L. Regano, D. Maiorca, G. Giacinto, and **M. Martalò**, “Evaluation of resource-aware HTTP/3 proxies for smuggling resilience in IoT environments,” *Proc. IEEE Global Commun. Conf. (Globecom)*, Taipei, Taiwan, December 2025.
- [C59] G. Pettorru, V. Pilloni, and **M. Martalò**, “Multilateration-assisted fingerprinting-based localization for dynamic IoT environments,” *Proc. IEEE Int. Mediterranean Conf. Communications and Networking (MeditCom)*, pp. 1-6, Nice, France, July 2025.
- [C58] G. Pettorru, A. Coni, V. Pilloni, and **M. Martalò**, “A reliability index for position estimation in trustworthy location-based services,” *Proc. IEEE Int. Mediterranean Conf. Communications and Networking (MeditCom)*, pp. 1-6, Nice, France, July 2025.
- [C57] G. Pettorru, G. Nurchis, V. Pilloni, and **M. Martalò**, “How do jamming attacks impact the performance of RSS-Based localization techniques?,” *Proc. IEEE Int. Conf. Commun. (ICC)*, pp. 2707-2712, Montreal, Canada, June 2025.
- [C56] C. Huidobro, M. Gutierrez Gaitán, C. Oberli, G. Pettorru, **M. Martalò**, and V. Pilloni, “Assessing the Interplay between IoT Localization Accuracy and the Two-Ray Channel” *Proc. IEEE Wireless Commun. Network. Conf. (WCNC)*, pp. 1-6, Milan, Italy, March 2025.
- [C55] **M. Martalò** and V. Pilloni, “A technology-agnostic framework for trustworthy crowdsourcing services in green 6G IoT,” *Proc. IEEE Global Commun. Conf. (Globecom)*, Cape Town, South Africa, December 2024.
- [C54] F. Marcello, G. Pettorru, **M. Martalò**, and V. Pilloni, “Preserving privacy in CSI-based human activity recognition: a data obfuscation case study,” *Proc. IEEE Global Commun. Conf. (Globecom)*, pp. 2822-2827, Cape Town, South Africa, December 2024.

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Cagliari, September 21, 2026

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